

Numerical Method

CSIT — 3rd Semester — 2080 — Answer Sheet
Subject Code: CSC212

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. How secant methods differs from Newton Raphson method? Derive the formula for Secant Method. Solve the equation $\cos x + 2 \sin x - x^2 = 0$ using Secant method. Assume error precision as 0.01. Discuss drawbacks of the Newton Raphson method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Define the terms interpolation and extrapolation. Write down algorithm and program for Newton's divided difference interpolation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How Gauss Jordan method differs from Gauss Elimination method? Solve the following system of equations using Gauss Jordan method. How can we use Gauss Jordan method to find inverse of a matrix? Discuss. $2x - y + 4z = 15$, $2x + 3y - 2z = 43$, $x + 2y - 4z = -4$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Define the terms approximate error and relative approximate error? Discuss the working of Half Interval method for finding roots of non-linear equation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

**5. Construct Newton's backward difference table for the given data points and approximate the value of $f(x)$ at $x=45$.
x: 45.10 20.30 40.50
f(x): 0.9850 0.9340 0.8660 0.7660 0.643**

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Fit the quadratic curve through the following data points and estimate the value of $f(x)$ at $x=2$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Factorize the following matrix using Cholesky method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. How can we calculate derivatives of discrete (tabulated) functions? Write down its algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Find following integral using composite trapezoidal rule for using 2 segments ($k=2$) and 4 segments ($k=4$).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Approximate the solution of $y' = 3x^2$, $y(1) = 1$ using Taylor's Series method using first four terms. Approximate the value of $y(2)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Solve the Poisson's equation $\nabla^2 f = xy$ with $f = 2$ on boundary by assuming square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ and $h = 1$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write down program for solving ordinary differential equation using Heun's method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.